



NITI Aayog



SECTOR AT A GLANCE 13

Energy Efficiency in India

Energy in India

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ENERGY EFFICIENCY IN INDIA

ENERGY IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Energy efficiency involves consuming less energy to achieve the same output while minimising waste. It encompasses various activities such as building energy efficiency, industrial energy management, appliance and equipment standards, transportation efficiency, process efficiency, agricultural efficiency, and energy recovery. It plays a pivotal role in India's sustainable development and energy security.

By prioritising energy efficiency initiatives, India can enhance energy security, reduce import dependency, and diversify its energy mix. The industrial sector, buildings, and transportation are the key areas where energy savings can be achieved.

However, challenges such as limited awareness and uptake of energy-efficient appliances, particularly among rural residents, constrained institutional capacity in energy efficiency cells and units, financing limitations for energy service companies, and the absence of systematic data on energy savings across sectors impede the widespread adoption of energy-efficient practices.

Given these opportunities and challenges, strengthening market mechanisms for energy efficiency and promoting energy-efficient technologies are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to the National Mission for Enhanced Energy Efficiency, Perform, Achieve, and Trade scheme, and UJALA scheme focus on creating marketing mechanisms to enable trading of excess energy savings. These efforts, along with several state- and local-level policies and initiatives, seek to reduce energy intensity of Indian economy by promoting energy efficiency and conservation across sectors.

Promoting energy efficiency across all major sectors of the economy, encouraging greater uptake among consumers, promoting capacity-building for state government officials and energy service companies, and improving state-level data capabilities. This note presents key trends related to different components of energy efficiency in India and actions for the way forward.

Figure 1: Key energy efficiency statistics

42 Mtoe*

Energy savings for 2020-21, i.e. 4.73% of the country's total primary energy supply

INR 152,241 cr.

Per annum monetary savings due to energy savings

267.98 MN tonnes

Annual reduction in CO₂ emissions due to energy savings

18.51%

Decrease in India's energy intensity from 2011-12 to 2020-21

45%

Reduction in emissions intensity of India's GDP targeted by 2030

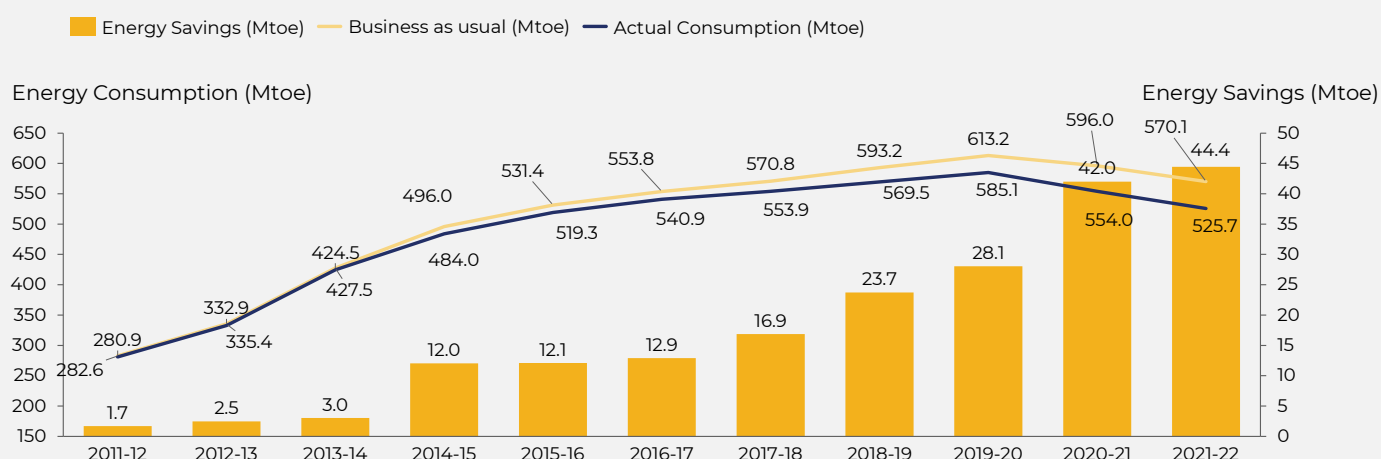
7 Mtoe

Energy savings aimed at by India by decreasing consumption within intensive industries

Source: Bureau of Energy Efficiency, Impact Assessment of Energy Efficiency for Year 2020-21, 2022; Union Cabinet chaired by the Prime Minister, India now stands committed to reduce Emissions Intensity of its GDP by 45 percent by 2030, 2022; Ministry of Power, Energy Efficiency, Accessed July 2023

Note: Mtoe refer to million tonnes of oil equivalent

Figure 2: Impact of various energy efficiency measures on India's energy consumption over the past decade



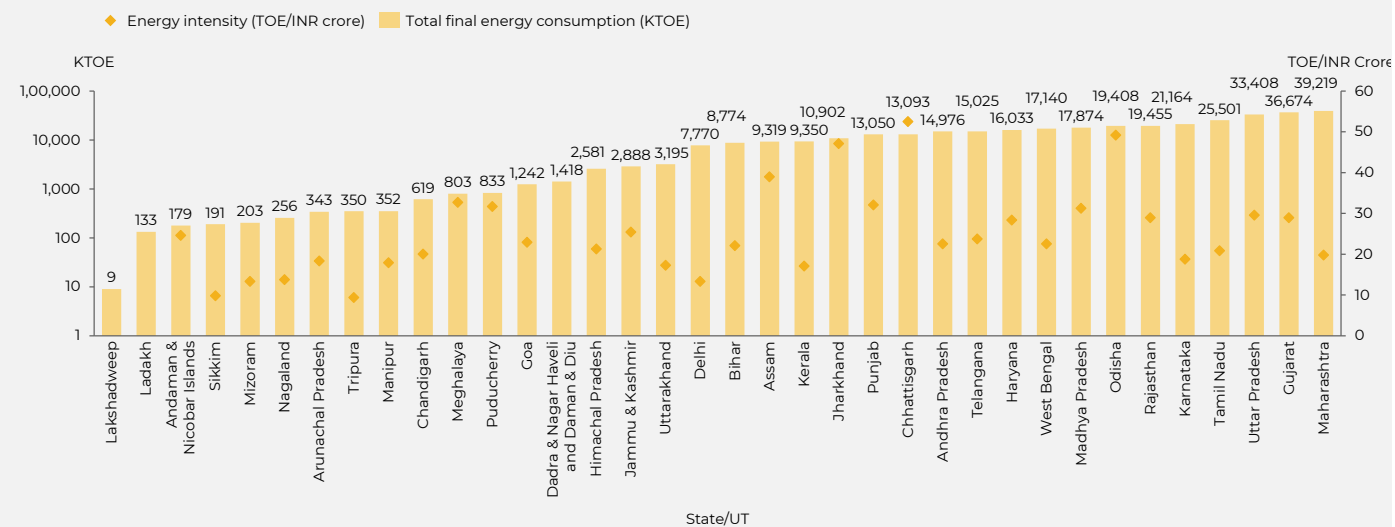
Source: Bureau of Energy Efficiency, Ministry of Power, Impact of Energy Efficiency Measures For The Year 2021-22, 2023

Figure 3: Sector-wise energy savings, 2021-22

Sector	Total energy saving (Mtoe)	Percentage share of energy saving (%)
Industry (includes savings from MSMEs and PAT exclusive of DISCOM, buildings, and railways)	24.3600	54.9%
Domestic (S&L and UJALA)	17.2200	38.8%
Transport (including railways)	2.0300	4.6%
Others (including municipal)	0.7300	1.6%
Buildings	0.0428	0.1%
Agriculture (including star rated pumps)	0.0060	0%
Total	44.43	100%

Source: Bureau of Energy Efficiency, Ministry of Power, Impact of Energy Efficiency Measures For The Year 2021-22, 2023

Figure 4: State-wise total final energy consumption (TFEC) and energy intensity (TFEC/unit state GDP), FY 2019-20, by increasing energy intensity



Source: Ministry of Power, State Energy Efficiency Index 2021-22, 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by increasing the adoption of energy efficiency practices across sectors. These primarily include overall reduction in energy demand, reduced emissions and enhanced energy security.



Reducing overall energy consumption:

As per Ministry of Power, in 2020-21, India consumed 581.6 Mtoe of energy and saved 42 Mtoe due to energy efficiency measures, which is about 7% of the total energy consumed. The percentage saving in total energy consumption due to energy efficiency measures has consistently increased from <0.5% in 2011 to 7% in 2021.¹ Continuing this process is critical to set the country on a sustainable growth pathway while reducing its carbon intensity.



Reducing emissions and meeting climate commitments:

By optimising energy use and reducing wastage, increasing energy efficiency often helps lower greenhouse gas emissions, mitigates the impact of fossil fuels, and lessens the strain on natural resources.² For example, as per the State Energy Efficiency Index, incorporating energy efficiency measures in municipal water supply systems can conserve water and energy, resulting in cost savings and improved service quality.³ Similarly, adopting climate-friendly infrastructure across the agricultural value chain, such as energy-efficient cold chain technologies and drip irrigation, can significantly reduce carbon emissions.⁴



Promoting new industries:

India's State Energy Efficiency Index highlights that energy efficiency programs are vital in facilitating the transition of high-energy consumption sectors towards sustainable solutions.⁵ This includes promoting the adoption of electric vehicles in the transportation sector, implementing sustainable building practices in the construction sector, transitioning to renewable energy-powered agriculture, and ensuring the adoption of energy efficiency practices in municipal services.



Improving economic growth and job creation

As the Indian economy expands, energy consumption continues to rise. However, strategic efforts to improve energy efficiency have mitigated this growth. Between 2010 and 2018, energy efficiency measures prevented an additional 12% of energy use, resulting in substantial cost savings for industries, businesses, and households, thereby enhancing economic growth.⁶ The deployment of energy-efficient technologies and practices can create employment opportunities in the manufacturing, construction, and services sectors. And investing in energy efficiency stimulates innovation and the development of new technologies, driving economic competitiveness.⁷



Driving innovation and technological development

1 Ministry of Power, National Electricity Plan, 2023

2 Ministry of Power, State Energy Efficiency Index, 2021-22

3 Ministry of Power, State Energy Efficiency Index, 2021-22

4 Ministry of Power, State Energy Efficiency Index, 2021-22

5 Ministry of Power, State Energy Efficiency Index, 2021-22

6 IEA, E4 Country Profile: Energy Efficiency in India, 2021

7 CEEW, Future Skills And Job Creation With Renewable Energy In India, 2019

Investing in energy efficiency stimulates innovation and drives the development of new technologies. India's energy efficiency programs have encouraged research and development in areas like smart grids, efficient appliances, and renewable energy integration. These innovations not only enhance energy performance but also position India as a global leader in sustainable solutions.⁸

Challenges

This section highlights some of the major challenges and constraints facing energy efficiency in India. Addressing these challenges requires a multi-faceted approach involving government policies, private sector initiatives, public awareness campaigns, and international cooperation. Encouraging investments in energy efficiency, providing incentives and subsidies, strengthening regulatory frameworks, promoting research and development, and fostering partnerships between governments, businesses, and civil society are essential steps toward realizing the full potential of energy efficiency in India. Such efforts will ensure reduced energy intensity and reduced emissions.



Inadequate awareness and uptake of energy-efficient appliances:

According to India Residential Energy Survey (IRES) 2020, the Bureau of Energy Efficiency's star label, which indicates the energy and cost-saving potential of a product, is known to only one out of four Indian households. This label is less familiar to rural residents, as only 19% are aware of it, compared to 38% of urban dwellers. Additionally, despite the widespread use of ceiling fans in 90% of homes, only 3% are energy efficient. This could be due to their higher price and limited availability in the market.⁹



Limited institutional capacity for implementing EE:

EE cells and unit offices responsible for collecting data on energy efficiency programs in different sectors have been established in just two states, Andhra Pradesh and Telangana.¹⁰ The State Energy Efficiency Index suggests that the State Designated Agencies (SDAs) need to collaborate closely with these units to facilitate the localised implementation of energy efficiency (EE) initiatives. It also highlights the need to conduct regular capacity-building programs to enhance government officials' knowledge and skills.



Constrained funding at the state level:

Only 2 states have incorporated ECBC in the state's bylaws and allocated funds to Third-Party Assessors (TPAs) and urban local bodies responsible for certifying and enforcing ECBC respectively. Further, just six State Designated Agencies (SDAs) financially support energy efficiency programs in their municipalities. Punjab is the only state with a dedicated budget for functioning Demand Side Management (DSM) cells that promote energy efficiency in agriculture and DISCOMS.¹¹



Insufficient financing for energy service companies (ESCOs):

Despite an estimated investment potential of INR 750 – 1,500 billion in energy efficiency (EE) projects, many end users are unable to implement these projects due to technical capacity limitations, financial constraints or inadequate focus and time for EE investments. ESCOs play a crucial role in overcoming critical barriers in the EE market by identifying opportunities, collaborating with equipment manufacturers, and managing EE technology. However, ESCOs struggle to secure finance from banks, possibly because of an absence of energy

⁸ IEA, Technology Innovation to Accelerate Energy Transitions, 2019

⁹ CEEW, Awareness and Adoption of Energy Efficiency in Indian Homes, 2020

¹⁰ Ministry of Power, State Energy Efficiency Index, 2021-22

¹¹ Ministry of Power, State Energy Efficiency Index, 2019

efficiency-focused guidelines on bank loans by the RBI. This lack of guidance hampers ESCOs' access to funds, impeding the growth of the ESCO industry.¹²



Inadequate data quality related to energy efficiency:

The India State Energy Efficiency Index 2019 reveals several key challenges pertaining to energy efficiency data. Firstly, there is a deficiency in systematically collecting data on energy savings across sectors, a lack of Monitoring and Verification (M&V) frameworks to establish baselines and track improvements, and insufficient practices for reporting and disclosing data, particularly regarding the energy consumption of public buildings. While a few states shared partial data on energy savings from specific buildings and efficient street lighting, none provided information on the energy intensity (passenger-km/litre) of State Road Transport Undertakings (SRTUs). Additionally, there is a notable absence of a structured process within State Designated Agencies (SDAs) to accurately and comprehensively capture energy savings.¹³



Limited state-level end-use energy data:

Adequate end-use energy data on state energy efficiency initiatives is not available with the SDAs. There is need for the SDAs to collaborate with relevant organisations to ensure regular updates on end-use energy data, energy intensity, and energy savings across various sectors.¹⁴ This could also enable setting quantifiable energy savings targets which most states do not have- which in turn hinders state-level action.

¹² Sarkar A. and Sundararajan S., Transforming India's energy efficiency market by unlocking the potential of private ESCOs, 2020; CEEW and National Resources Defense Council (NRDC), Greening India's Financial Market: Investigating Opportunities for a Green Bank in India, 2016; Small Industries Development Bank of India (SIDBI), REol for Partial Risk Sharing Facility for Energy Efficiency Project, 2020

¹³ Ministry of Power, State Energy Efficiency Index, 2019

¹⁴ Bureau of Energy Efficiency, State Energy Efficiency Index, 2020

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee energy efficiency.



NITI Aayog

The NITI Aayog serves as the apex public policy think tank of the Government of India. Within the energy vertical, deals with five key sectors: power, coal, petroleum and natural gas, new and renewable energy, and atomic power. Within the energy efficiency domain, NITI Aayog conducts comprehensive policy analysis and strategic planning to promote the adoption of energy-efficient practices and technologies.



Ministry of Power

The ministry administers, formulates, and implements power sector-related policies and provides R&D and technical assistance relating to grid integration, transmission system networks and distribution systems in the States/UTs.



Ministry of New and Renewable Energy (MNRE)

The ministry develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security.



Bureau of Energy Efficiency

The bureau is the nodal authority for all GH2 projects in India, assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act.



Indian Renewable Energy Development Agency

The agency finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies.



Small Industries Development Bank of India (SIDBI):

SIDBI plays a crucial role in promoting energy efficiency among small and medium enterprises (SMEs). It offers financial assistance and advisory services to SMEs for implementing energy-efficient technologies and practices. SIDBI also collaborate with technical institutions to offer capacity building programs for SMEs on energy management and conservation.



Power Finance Corporation (PFC)

A public sector financial institution under the Ministry of Power, PFC finances the power sector through loans and investments. They play a crucial role in funding generation, transmission, and distribution projects, promoting sector growth.

Major policies and programs

This section highlights existing policies related to integrated energy assessments and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Energy Conservation Act, 2001:

The Energy Conservation Act 2001 aims to promote energy efficiency and reduce energy consumption. It established the Bureau of Energy Efficiency (BEE) as a statutory body under the Ministry of Power, responsible for implementing the act's provisions.¹⁵



National Mission for Enhanced Energy Efficiency (NMEEE):

This is one among the eight national missions as under the National Action Plan on Climate Change (NAPCC). It is aimed at strengthening the market for energy efficiency through the implementation of innovative business models in the energy efficiency sector. It consists of four initiatives to enhance energy efficiency in energy-intensive industries.



Perform, Achieve, and Trade (PAT) Scheme:

The scheme is aimed at reducing Specific Energy Consumption (SEC) in energy-intensive industries. To achieve this, a market-based mechanism is employed allowing certification of excess energy savings which can be traded, thereby enhancing cost-effectiveness. Implementation of PAT cycle-II from 2016-19 resulted in total energy savings of about 14.08 million tonnes of oil equivalent (MTOE), translating into avoiding about 68 million tonnes of CO₂ emission.¹⁶



Market Transformation for Energy Efficiency (MTEE):

MTEE aims to expedite the shift to energy-efficient appliances in select sectors through innovative business models and incentives. Similarly, the Bachat Lamp Yojana of MTEE developed for replacing inefficient bulbs with Compact Fluorescent Lamps (CFLs) led to energy savings of about 4 billion units/year.¹⁷



Unnat Jyoti by Affordable LEDs for All (UJALA):

UJALA was launched by the Ministry of Power to provide energy-efficient LED bulbs to domestic consumers at an affordable price. Over 360 million LEDs have been distributed nationwide, leading to an annual energy saving of 47,784 million units of electricity, a reduction in peak demand by 9,566 MW, and a yearly decrease in CO₂ emissions by 38.7 million tonnes.¹⁸



Energy Conservation Building Code (ECBC):

The Ministry of Power initiated the Energy Conservation Building Code (ECBC) with the aim of fostering energy efficiency within the commercial building industry. It sets minimum energy standards for new buildings with a 100 kilowatt (kW) connected load or contract demand of 120-volt amps kVA or more. The Energy Conservation Building Code (ECBC) includes three levels of energy performance standards: ECBC, ECBC Plus, and Super ECBC. In ascending order,

¹⁵ Bureau of Energy Efficiency, The Energy Conservation Act, 2001, accessed July 2023

¹⁶ Bureau of Energy Efficiency, Perform Achieve Trade, accessed July 2023

¹⁷ Ministry of Power, Status of Implementation of National Mission for Enhanced Energy Efficiency (NMEEE), 2022

¹⁸ Ministry of Power, Salient features of UJALA and SLNP programmes, 2022

they offer savings of about 25%, 35%, and 50%, respectively, compared to conventional buildings.¹⁹



Unnat Takniki Pradarshan Kendra (UTPRERAK):

UTPRERAK, also known as Advanced Industrial Technology Demonstration Centre (AITDC) has been set up by BEE to showcase and demonstrate energy-efficient technologies across key industry sectors. This center will facilitate knowledge exchange through workshops and seminars, allowing industry professionals to share best practices from various key sectors. It will also serve as a strategic capacity-building institution, offering comprehensive training and education in energy efficiency. Over the next five years, it aims to provide intensive training to over 10,000 energy professionals from the industry and other relevant sectors across India.²⁰



Standards and Labelling (S&L) scheme:

The S&L scheme was launched by the Ministry of Power to induce a reduction in appliance energy consumption without diminishing the services it provides to consumers. Covering star labelling for 24 appliances currently, the scheme prescribes minimum energy performance levels for appliances/ equipment on a scale of 1 to 5. In 2017-18 alone, this programme resulted in electricity savings of 48.46 BU, amounting to emission reduction worth 40.03 million tons. The success of this scheme paved way for the development of India's Cooling Action Plan (2019) as well as new star-labelling programmes for commercial air-conditioners and solar panels.

Framework for Energy Efficient Economic Development (FEEED):

Launched by the Bureau of Energy Efficiency (BEE), FEEED aims to facilitate financial institutions' participation in promoting energy efficiency projects. This framework establishes financial instruments to encourage banks and other institutions to provide loans for energy-saving upgrades in industries, large commercial establishments, and other sectors. By enabling easier access to financing, FEEED aims to accelerate the adoption of energy-efficient technologies and practices.



Demand Side Management (DSM) Scheme:

This scheme includes multiple initiatives targeting energy efficiency across sectors of the Indian economy. These include Agriculture DSM (focuses on creating a market-based framework for implementing pilot projects on energy-efficient pumpsets (EEPS) and raising awareness among stakeholders for their adoption), Municipal DSM (focuses on improving the overall energy efficiency of Urban Local Bodies (ULBs) to achieve substantial electricity consumption savings and cost reduction), Capacity Building of DISCOMS, Energy Efficiency in Small and Medium Enterprises sector, and the Strengthening Institutional Capacity of States initiative. By focusing on demand-side interventions, DSM aims to optimize energy usage, enhance efficiency, and promote sustainable energy practices across diverse sectors of the economy.²¹



¹⁹ Bureau of Energy Efficiency, ECBC Commercial, July 2023

²⁰ Ministry of Power, Power ministry establishes UTPRERAK, a Centre of Excellence to Accelerate Adoption of Energy Efficient Technologies in Indian Industry, 2023

²¹ Ministry of Power, Schemes to Promote Energy Conservation and Energy Efficiency, accessed 2024

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the NITI Aayog and the Ministry of New and Renewable Energy.



Improving energy efficiency in key sectors

Promote energy efficiency in industries

- Strengthen the implementation of the Perform, Achieve, and Trade (PAT) scheme by the Bureau of Energy Efficiency (BEE). This scheme mandates energy-intensive industries to achieve specific energy intensity reduction targets. Enhance monitoring, compliance, and enforcement mechanisms.
- Promote energy-efficient technologies and practices in industrial clusters through sector-specific programs. Collaborate with industry associations, research institutions, and technology providers to disseminate best practices.
- Establish a robust data collection system to monitor energy consumption trends, identify inefficiencies, and track progress. Encourage industries to adopt energy management systems (ISO 50001) for continuous improvement.

Promote energy efficiency in agriculture

- Implement demand-side management (DSM) programs for agricultural pump sets. Replace inefficient pumps with energy-efficient ones and promote solar-powered pumps.
- Train farmers on optimal irrigation practices, crop selection, and energy-efficient agri-processing techniques. Collaborate with agricultural universities and extension services.

Promote energy efficiency in buildings

- Strengthen the Energy Conservation Building Codes (ECBC) by incorporating stricter standards for new and existing buildings. Encourage states to adopt and enforce ECBC rigorously.
- Promote green building certification (such as GRIHA and LEED) to incentivize energy-efficient design, construction, and operation. Provide financial incentives and expedited approvals for certified projects.
- Launch awareness campaigns targeting builders, architects, and consumers to highlight the economic and environmental benefits of energy-efficient buildings.



Enhancing uptake and awareness of energy-efficient appliances

- Expand the UJALA scheme (Unnat Jyoti by Affordable leds for All) to include other energy-efficient appliances (fans, refrigerators, air conditioners). Provide consumer subsidies and create awareness through targeted campaigns.
- Collaborate with appliance manufacturers to develop and market energy-efficient products. Introduce labeling programs (like BEE's star rating) for appliances beyond lighting.

	- Engage with consumer forums, housing societies, and schools to educate citizens about the benefits of energy-efficient appliances.
 Promote capacity-building for state officials and Energy Service Companies (ESCOs)	- Establish training programs for state energy officials on energy efficiency policies, financing mechanisms, and project implementation. Foster knowledge exchange among states. - Strengthen ESCOs' capacity to implement energy efficiency projects. Encourage public-private partnerships (PPP) for ESCO-led initiatives.
 Improving state-level energy efficiency and end-use data	- Invest in robust data collection systems, including smart meters, to track energy consumption patterns at the state level. Use data analytics for evidence-based decision-making. - Collaborate with research institutions (both domestic and global) to conduct state-level energy audits and identify sector-specific opportunities for improvement.

Recommended readings

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